

Proposed cross-city PM_{2.5} transfer-learning framework

Data & preprocessing

Instruments

CPCB CAAQM network
PM_{2.5} (BAM) + AT, RH,
WS, WD · 3-hourly
Delhi 40 / Kol 10 / Guw 4

Preprocessing

- IDW + Kalman gap-fill
- 14 engineered features
- climatology-residual target
- 70/15/15 interleaved split

k-NN graph, $k = 3$
Gaussian weights, $\sigma = 5$ km

Stage 0
source-only pre-train

**Shared inductive
ST-GNN encoder**
TCN → GAT×2 → TCN

~25k parameters,
no $|V|$ dependence (θ)

Cross-city transfer

Stage 1 — PT-FT

load θ → fine-tune
on target d% (MSE)

3-way verify

zero-shot ·
scratch · transfer

Stage 2 — Graph-DANN

adversarial align of
mean+max readout
(GRL + 3-city disc.) → FT

Evaluate

R^2 · MAE ·
RMSE · MAPE